

# Sage Bot

## Features

- Fully autonomous swarm robot kit with 89V51RD2-based ISP programmable development platform
- Optional 2.4GHz wireless modules for wireless communication
- Capability for multiple robots to communicate with each other to perform a task (swarm robotics)
- Five sensor IR modules for grid solving and line following applications
- Dual TSOP obstacle-detection modules
- Comes with 9V, 220mAh NiMH battery with charger for the bot



**Price: Rs 3500**

with a microcontroller, the robot can be programmed and made autonomous. More advanced sensors and high-performance processors also help. If the robot can be connected to a computer through wire or wirelessly, there can be a great increment in the level of autonomy.

"Fully autonomous robots are available in India. But as more complex features such as vision and self bal-

ancing are added, the cost increases," informs Syed Ross Farooq, managing partner, Robosoft Systems.

## Trends world over

For basic robot kits, trends are the same outside India. However, in advanced robotics, India is still far behind the developed nations. In some countries, many advanced sensors like laser measurement sensors and lidars

are in use. These sensors gather a large amount of data, for which very powerful microcontroller or computers are used. However, their cost is a limiting factor. An LMS starts at \$2000 and goes up to \$8000 for advanced versions.

Worldwide, the focus is on implementing more and more technologies into robotics. Complex computational and processing techniques are used in robots to make them behave as humans. Advances in digital image processing techniques like image normalisation using Eigen arithmetic, image acquisition toolbox, Eigen algorithm, discrete cosine transform compression, neural network, fuzzy logic and Kalman filtering, and digital audio processing techniques like Melcept frequency calculation, K-Mean clustering and text-to-speech conversion make this possible. Also, the use of higher-end microcontrollers with high-speed processing and accuracy makes calculations more precise than ever.

## How to buy?

If you are buying the kit for some specific project or personal and hobby use, go for an all-in-one kit which has many sensors, motors, display, input and output modules, PC connectivity and easy programming.

If you do higher level of research on robotics, you need a robot which can carry high payloads, connect with a variety of sensors, and has lots of expansion options, PC connectivity, complex remote controls and wireless connectivity, vision processing, etc. These kits usually cost above Rs 15,000 and are bought by universities, government organisations or private organisations for R&D purposes.

Jayakrishnan T., director (India), Energid Technologies, suggests to go for a kit which is modular, reconfigurable and reusable for different applications. The additional expenses can be offset by multiple use of the kit. Going for a low cost, on the other hand, would mean compromising the quality or functionality to reduce the initial expense apart from using it for a

# Universal Robotics Development Platform

## Features

- A single platform for developing different kinds of educational robots, ranging from simple line-following robot to advanced speech-controlled robot and computer-controlled robot
- Accompanied with around 15 different sensors modules, combining which allows one to create 20+ different machines
- Different mechanical parts can be added to make pick-and-place robot and more
- Choose from 8051, AVR, Arduino and PIC for the controller—the main unit
- Add different sensor modules and change programming of the controller to provide different functions to the machine



**Price: Rs 4000**